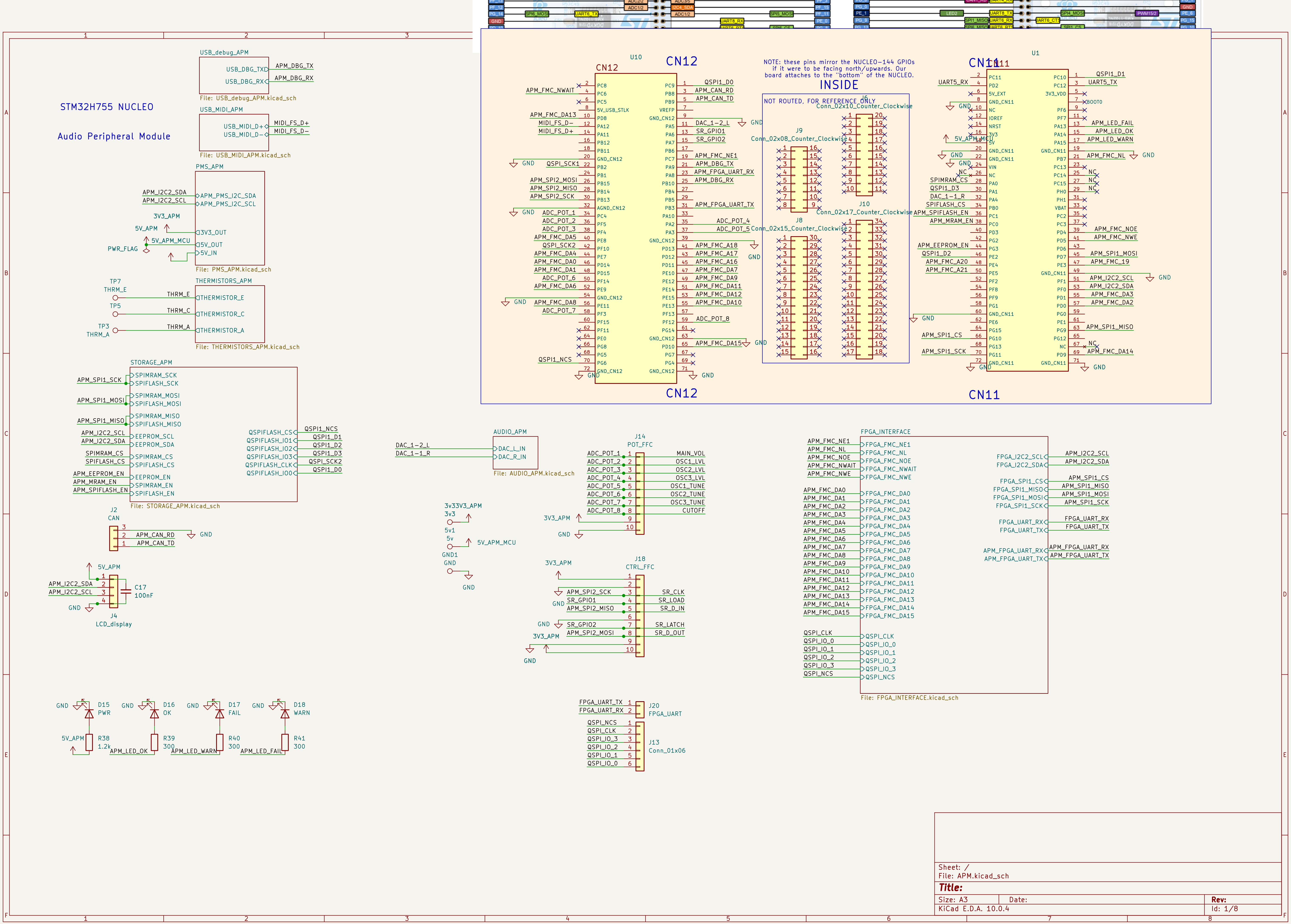
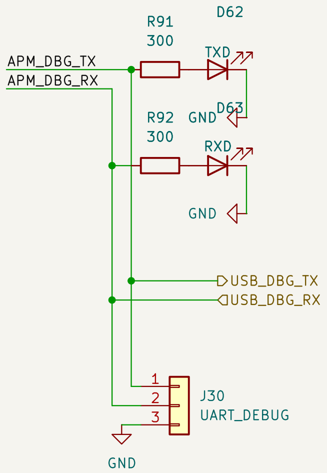


STM32H755 NUCLEO
Audio Peripheral Module
USB_debug_APM
File: USB_debug_APM.kicad_sch
USB_MIDI_APM
File: USB_MIDI_APM.kicad_sch
PMS_APM
File: PMS_APM.kicad_sch
THERMISTORS_APM
File: THERMISTORS_APM.kicad_sch
STORAGE_APM
File: STORAGE_APM.kicad_sch
AUDIO_APM
File: AUDIO_APM.kicad_sch
FPGA_INTERFACE
File: FPGA_INTERFACE.kicad_sch
CN12
CN11
INSIDE
NOTE: these pins mirror the NUCLEO-144 GPIOs if it were to be facing north/upwards. Our board attaches to the "bottom" of the NUCLEO.
NOT ROUTED, FOR REFERENCE ONLY
J14 POT_FFC
J18 CTRL_FFC
J20 FPGA_UART
J13 Conn_01x06
J2 CAN
J4 LCD_display
D15 PWR
D16 OK
D17 FAIL
D18 WARN
Sheet: /
File: APM.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/8

UART -> USB Debug Port

UART debug messages avaiable via a
USB port and a 3 pin jumper for
TX, RX, and GND UART connections

USB2.0 DBG PORT



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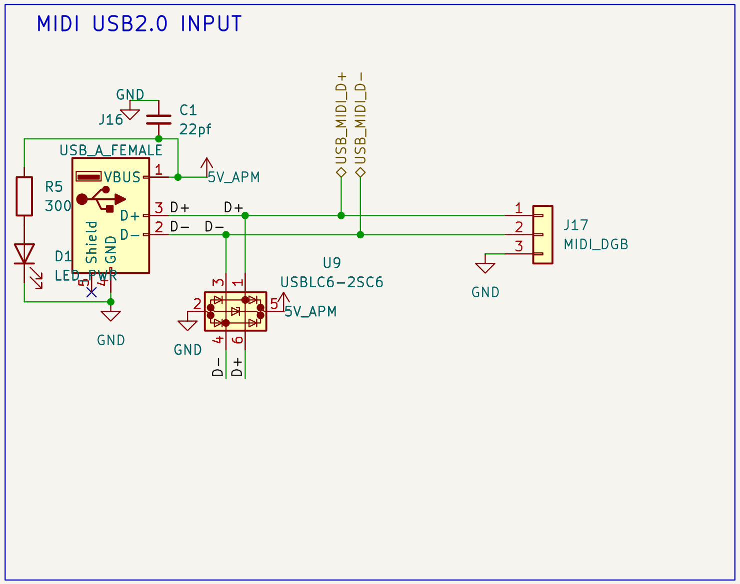
KiCad E.D.A. 10.0.4		Id: 2/8



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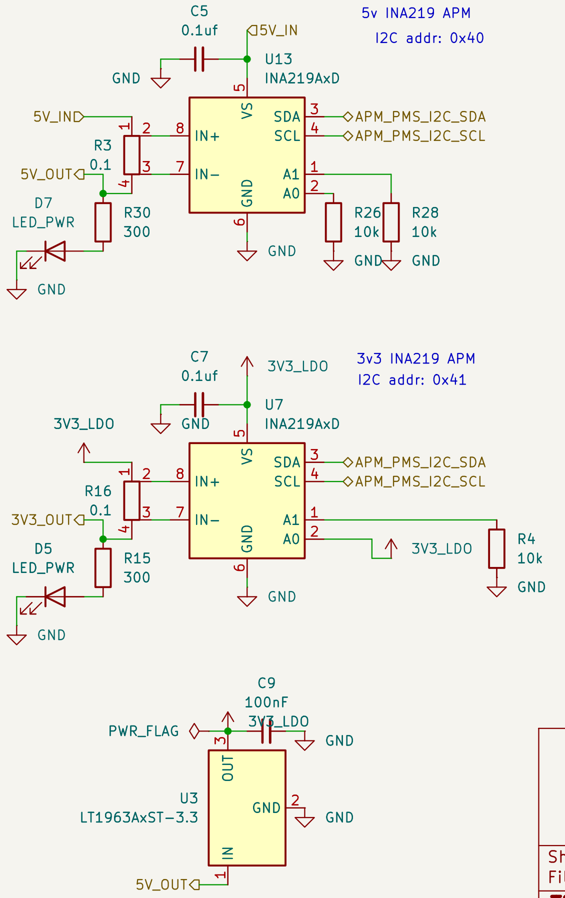
File: USB_MIDI_APM.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/8

Power Monitor and Supply

PMS has an INA219 current sensor
for each board/node in the
PCB. Each takes in the 5V from it's
parent board and supplies 5V to said
board's circuit (its peripherals)
thus enabling monitoring



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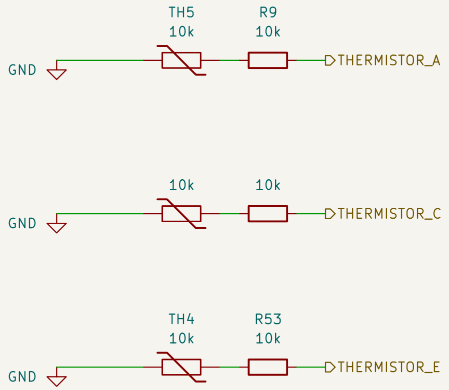
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 4/8



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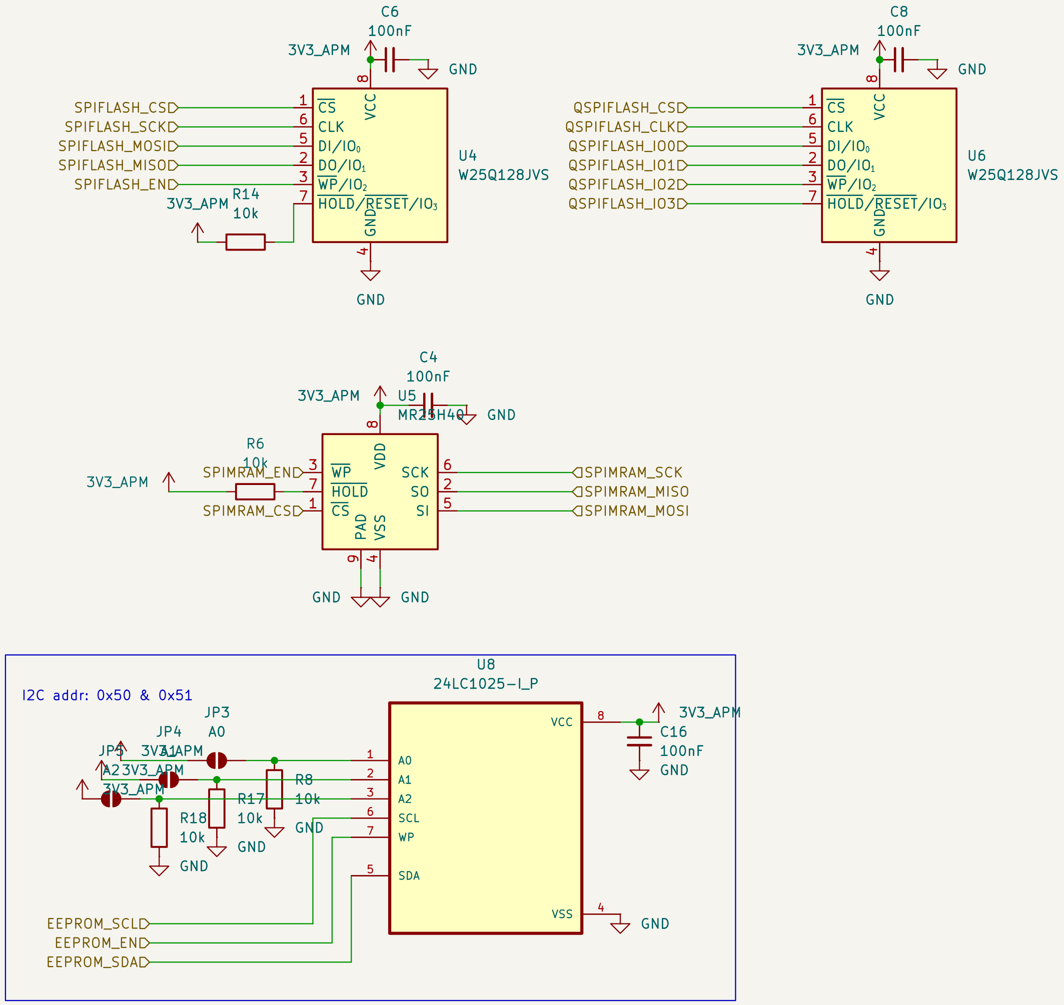
File: THERMISTORS_APM.kicad_sch

Title:

Size: A4 Date: **Rev:**

KiCad E.D.A. 10.0.4 Id: 5/8

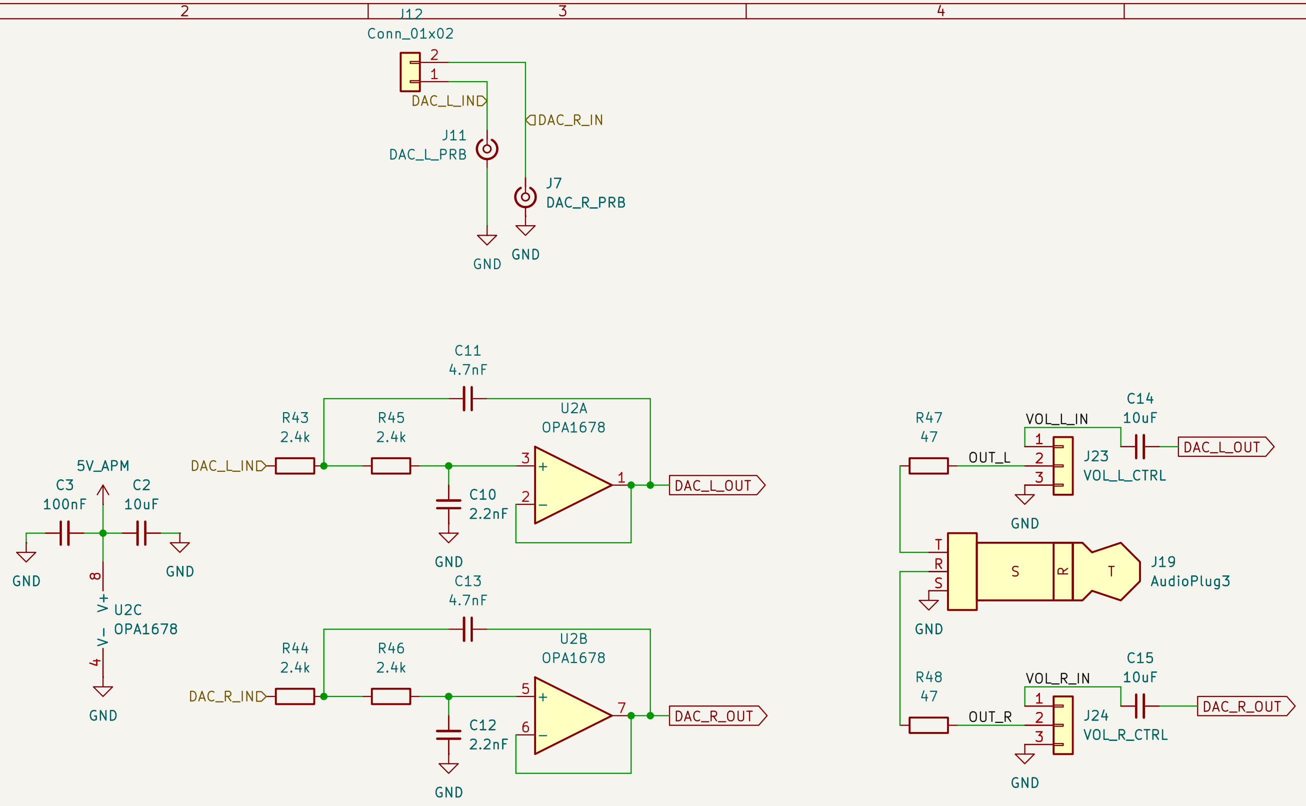


Sheet: /STORAGE_APM/
File: STORAGE_APM.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 6/8



Sheet: /AUDIO_APM/

File: AUDIO_APM.kicad_sch

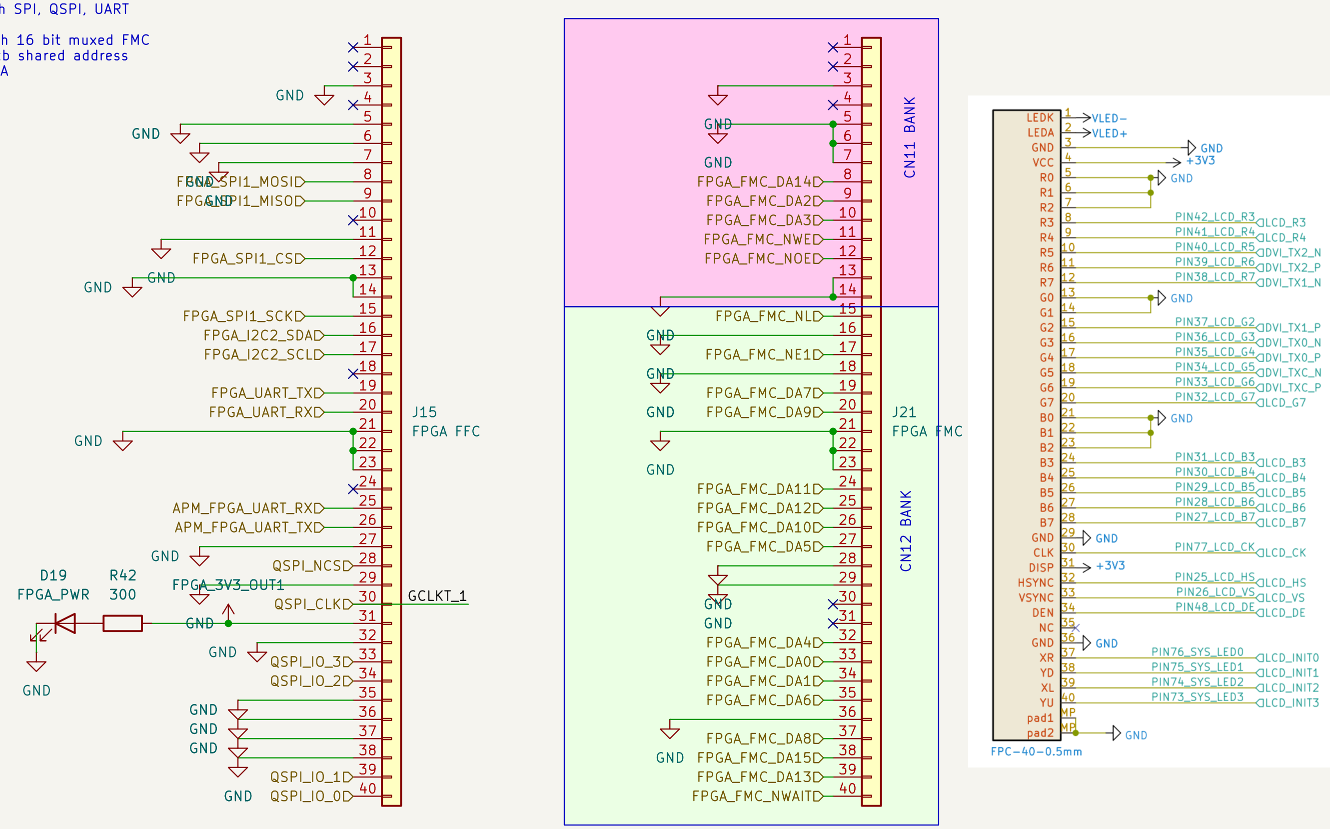
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 7/8

Dual FPC STM32<->Tang Nano 20K FPGA connection configuration. The STM32 side has two connectors where only one can be active at a time

- Configuration 1: FPGA_FFC with SPI, QSPI, UART connections.
- Configuration 2: FPGA_FMC with 16 bit muxed FMC configuration resulting in a 128kb shared address space between the STM32 & FPGA



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File: FPGA_INTERFACE.kicad_sch

Title:

Size: A4 Date:
KiCad E.D.A. 10.0.4

Rev:
Id: 8/8